CASE STUDY

Big Data in Agriculture

How data helps farmers grow more with less

What Is Big Data in Farming?

Huge amounts of information collected from farms and analyzed to help farmers make better decisions.

Field sensors

Measure soil moisture and nutrients

Drones and satellites

Show crop health from above

Weather and machines

Rain forecasts and tractor data

How It Works

1

Collect

Sensors, drones and machines gather data every day

2

Analyze

Software finds patterns and makes predictions

3

Act

Farmers water, plant and harvest at the best time

Case: A Smart Wheat Farm

Problem	Uneven rain and rising fertilizer costs made yields hard to predict.
Solution	Soil sensors and satellite maps showed when and where to water and fertilize.
Result	Less water and fertilizer wasted, healthier crops and steadier harvests.

Illustrative example based on common precision-farming practices

Benefits and Challenges

Benefits

- Higher crop yields
- Less water and fertilizer used
- Early pest and disease alerts
- Better planning

Challenges

- High cost of equipment
- Weak internet in rural areas
- Data privacy and ownership
- Skills and training needed

DISCUSSION QUESTION 1 OF 5

Which farm data is the most valuable?

A Soil moisture and nutrients

B Weather forecasts

C Crop images from drones and satellites

Pick one, then explain why. Or share your own idea.

DISCUSSION QUESTION 2 OF 5

Who should own the data collected on a farm?

A The farmer

B The technology company

C Both share it

Pick one, then explain why. Or share your own idea.

DISCUSSION QUESTION 3 OF 5

How can small farms benefit from big data?

A Share equipment with neighbors

B Use free phone apps and weather data

C Join a farming cooperative

Pick one, then explain why. Or share your own idea.

DISCUSSION QUESTION 4 OF 5

What is the biggest risk of relying on technology?

A Equipment breaks or the internet goes down

B Farm data is misused or sold

C Farmers lose traditional knowledge

Pick one, then explain why. Or share your own idea.

DISCUSSION QUESTION 5 OF 5

How can big data help farms with climate change?

A Save water with smart irrigation

B Predict extreme weather earlier

C Choose crops that suit the new climate

Pick one, then explain why. Or share your own idea.